

Jinpeng Yu

Homepage: jinpeng-yu.github.io ◇ Google Scholar ◇ Github

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Bios

I earned my Master's degree from ShanghaiTech University in July 2024, under the supervision of Prof. Shenghua Gao (HKU). Prior to that, I graduated with honors from the Harbin Institute of Technology (HIT) in June 2021. My current research interests include controllable image generation and multimodal understanding and generation. Additionally, I am also exploring embodied AI research.

Education

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| ShanghaiTech University | July 2024 |
| M.S. in Computer Science, supervisor: Prof. Shenghua Gao
<i>awarded Academic Excellence Scholarship for three consecutive years</i> | |
| Harbin Institute of Technology (HIT) | June 2021 |
| B.E. in Computer Science
<i>Top 5%, awarded Outstanding Graduate</i> | |

Publications

- Pseudo-plane Regularized Signed Distance Field for Neural Indoor Scene Reconstruction**
Jinpeng Yu^{*}, Jing Li^{*}, Ruoyu Wang, Shenghua Gao
International Journal of Computer Vision (IJCV), 2024
- GeoFormer: Learning Point Cloud Completion with Tri-Plane Integrated Transformer**
Jinpeng Yu, Binbin Huang, Yuxuan Zhang, Huaxia Li, Xu Tang, and Shenghua Gao
ACM International Conference on Multimedia (ACM MM), 2024
- SSR-Encoder: Encoding Selective Subject Representation for Subject-Driven Generation**
Yuxuan Zhang, Jiaming Liu, Yiren Song, Jinpeng Yu, Rui Wang, Hao Tang, Huaxia Li, Xu Tang, Yao Hu, Han Pan and Zhongliang Jing
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024
- GeoFormer: Learning Point Cloud Completion with Tri-Plane Integrated Transformer**
Yuxuan Zhang, Yiren Song, Jinpeng Yu, Han Pan and Zhongliang Jing
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024
- P²SDF for Neural Indoor Scene Reconstruction**
Jing Li^{*}, Jinpeng Yu^{*}, Ruoyu Wang, Zhengxin Li, Zhengyu Zhang, Lina Cao, and Shenghua Gao
arXiv preprint (arXiv), 2023

Projects

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|---|----------|
| Advanced Outpainting with Controllable Subject Brightness | Oct 2024 |
| <i>Recent Project, launch of Product AI-Tab</i>
Collected 2 million high-quality photographic datasets, extracted the corresponding saliency masks, and adjusted the subject brightness to implement a foreground perturbation strategy. | |
| Exploring InstantID Reconstruction with ControlNet-XS | May 2024 |
| <i>Internship Project, launch of Product Jinggou</i>
Utilized ControlNet-XS to replace the control architecture of InstantID, reduce inference resource consumption, and improve the model's effectiveness and efficiency. | |
| Crowd Counting and Localization based on Independent Instance Maps | May 2021 |
| <i>Bachelor's Thesis</i>
Re-implemented RDNet using independent instance maps to guide density map regression. | |

Services

Conference Reviewer

CVPR (2024, 2025), ECCV (2024), ACM MM (2024), ICME (2024)

Experiences

AGI Researcher at Xiaohongshu AIGC Team (Full-time Employee)	07/2024 - present
AGI Research Intern at Xiaohongshu AIGC Team	05/2023 - 06/2024
TA of Academic Paper Writing (SI200), Prof: Song Liu, ShanghaiTech	Autumn 2022
TA of Computer Vision II (CS272), Prof: Shenghua Gao, ShanghaiTech	Autumn 2021

Achievements

Academic Excellence Scholarship, awarded by ShanghaiTech	2022, 2023, 2024
Outstanding Graduate, awarded by HIT	June 2021
National Scholarship, awarded by HIT, Ministry of Education	2020
Outstanding Student, awarded by HIT	2019, 2020
Merit Student, awarded by HIT	2019, 2020

Skills/Hobbies

Programming	Python, Pytorch, C/C++
Hobbies	Hiking, Traveling and Photography